



Strongly Connected Components and Topological Sort.

1 Topological Sort

Definition 1. (connectivity)

1. Let $G = (V, E)$ be a non-directed graph. A **connected component** of G is a subset $U \subseteq V$ of maximal size in which there exists a path between every two vertices.
2. A non-directed graph G is said to be a **connected** graph if it only has one connected component.
3. Let $G = (V, E)$ be a directed graph. A **strongly connected component** of G is a subset $U \subseteq V$ of maximal size in which for any pair of vertices $u, v \in U$ there exist both directed path from u to v and a directed path from v to u .

1.1 Depth First Search (DFS)

As its name implies, depth-first search searches "deeper" in the graph whenever possible. Depth-first search explores edges out of the most recently discovered vertex v that still has unexplored edges leaving it. Once all of v 's edges have been explored, the search "backtracks" to explore edges leaving the vertex from which v was discovered. This process continues until all vertices that are reachable from the original source vertex have been discovered. If any undiscovered vertices remain, then depth-first search selects one of them as a new source, repeating the search from that source. The algorithm repeats this entire process until it has discovered every vertex.

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1 DFS(  $G$  ):
2   for  $v \in V$  do
3      $v.\text{visited} \leftarrow \text{False}$ 
4   end
5    $\text{time} \leftarrow 1$ 
6   for  $v \in V$  do
7     if not  $v.\text{visited}$  then
8        $\pi(v) \leftarrow \text{null}$ 
9       Explore(  $G, v$  )
10    end
11  end

1 Explore( $G, v$ ):
2  Previsit( $v$ )
3  for  $(v, u) \in E$  do
4    if not  $u.\text{visited}$  then
5       $\pi(u) \leftarrow v$ 
6      Explore(  $G, u$  )
7    end
8  end
9  Postvisit( $v$ )

1 Previsit( $v$ ):
2   $\text{pre}(v) \leftarrow \text{time}$ 
3   $\text{time} \leftarrow \text{time} + 1$ 

1 Postvisit( $v$ ):
2   $\text{post}(v) \leftarrow \text{time}$ 
3   $\text{time} \leftarrow \text{time} + 1$ 

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Properties of depth-first search. Depth-first search yields valuable information about the structure of a graph. Perhaps the most basic property of depth-first search is that the predecessor subgraph G_π does indeed form a forest of trees since the structure of the depth-first trees exactly mirrors the structure of recursive calls of explore-function. That is, $u = \pi(v)$ if and only if $\text{explore}(G, v)$ was called during a search of u 's adjacency list.

Additionally, vertex v is a descendant of vertex u in the depth-first forest if and only if v is discovered during the time in which $\text{pre}(u)$ has already been set but $\text{post}(u)$ hasn't¹. Another important property of depth-first search is that discovery and finish times have a parenthesis structure. If the explore procedure were to print a left parenthesis " u " when it discovers vertex u and to print a right parenthesis " u " when it finishes u , then the printed expression would be well-formed in the sense that the parentheses are properly nested.

The following theorem provides another way to characterize the parenthesis structure.

Theorem 1 (Parenthesis theorem). *In any depth-first search of a (directed or undirected) graph $G = (V, E)$, for any two vertices u and v , exactly one of the following three conditions holds:*

1. *the intervals $[\text{pre}(u), \text{post}(u)]$ and $[\text{pre}(v), \text{post}(v)]$ are entirely disjoint, and neither u nor v is a descendant of the other in the depth-first forest.*
2. *the interval $[\text{pre}(u), \text{post}(u)]$ is contained entirely within the interval $[\text{pre}(v), \text{post}(v)]$, and u is a descendant of v in a depth-first tree, or*

¹In terms of the presentation from last week, when u 's color is gray

3. the interval $[pre(v), post(v)]$ is contained entirely within the interval $[pre(u), post(u)]$, and v is a descendant of u in a depth-first tree.

Let us use the following notation: we say that line (x) of the recursive call 'Explore(G, v)' is executed at time t if at time t the running of the algorithm got to line number x of Explore where the vertex argument is v . Notice that since Explore is called **only once** on each vertex, the notation is well defined.

Proof. It's enough to prove that for any pair of vertices v, u , the inequality $pre(v) < pre(u) < post(v)$ implies $pre(v) < pre(u) < post(u) < post(v)$.

Consider the call 'Explore(G, v)'. Since the algorithm sets the value of $post(v)$ at line number (9), and it makes all of its recursive calls between lines (3)-(8), we have that the algorithm returns from its recursive calls before time $post(v)$. Furthermore, any line that has been executed during times $[pre(v), post(v)]$ is either a line of the call 'Explore(G, v)' or a line of one of the recursive calls.

In particular, we have that the line which sets $pre(u)$ is executed during times $[pre(v), post(v)]$, and therefore as part of a recursive call. In addition, because the recursive call was returned during the time interval, the assignment of $post(u)$ is made before $post(v)$. Thus, we have both $pre(v) < pre(u) < post(u) < post(v)$ and that u is a descendant of v in π ².

□

Corollary 1. Vertex v is a proper descendant of vertex u in the depth-first forest for a (directed or undirected) graph G if and only if $pre(u) < pre(v) < post(v) < post(u)$.

2 DFS on DAGS

the pre is well calibrated on DAGS.

Claim 1. Let $G = (V, E)$ be a graph, and let $v, u \in V$ such $u \rightarrow v$ (There is a path from u to v in G). And let π be a forest returned from the call **DFS**(G), If $pre(u) < pre(v)$ then v is a descendant of u in G .

Proof. By induction on the graph structure. Consider the pre order on the vertices, $V = v_1, v_2, \dots, v_n$, and the graphs seria $G_1 = \{u\}$ and $G_{i+1} = G_i \cup \{v_i\}$. Denote also by π_i the recursive tree returned by the application of the algorithm on G_i . We will prove by induction on i that the claim hold for every u, v in G_i , namely that if $pre(u) < pre(v) \leq i$, then v is a descendent of u in π_i .

1. Base, Consider G_1 and one can shoe by bruteforce that the claim holds.
2. Assumption. Assume the correctness of the claim for any $i' < i$.
3. Step. Consider vertices v, u such $u \rightarrow v$, in G_i and that $pre(u) < pre(v)$. Since the running of the algorithm over G_i preceding the explore call of v_i is identical to the running over G_{i-1} , we get that $\pi_{i-1} \subset \pi_i$ and it is left to

²The algorithm defines π to be the recursive tree

prove where $v = v_i$. Let v' be predecessor to v on that path, namely the last vertex on the ' $u \rightarrow v$ ' path before v .

- If $v' = v_j$ for $j < i$, then consider the call ' $\text{Explore}(G_i, v_j)$ ', since $\text{pre}(v_{i-1}) = i - 1 < \text{pre}(v_i)$, then at the beginning of the recursive call v_i has not yet visited and the algorithm sets $\pi(v_i) \leftarrow v_{i-1}$, now using the induction assumption, v_{i-1} is a descendant of u in $\pi_{i-1} \subset \pi_i$ we have also that v_i is a descendant of u in π_i .
- Else, $v' = v_j$ for $j > i$. That means that the algorithm got to v through the iteration loop in the DFS routine, and not through a sibling vertex. Therefore any of the paths between u to v , there is a vertex w on that path such $w = v_j$ for $j < i$. Take w to be the minimal according to the pre order. And consider the recursive call ' $\text{Explore}(G_i, w)$ '. Since Denote by w' be its successor on that path.

□

Claim 2. Let $G = (V, E)$ be a DAG, and let $v, u \in V$ such $u \rightarrow v$ (There is a path from u to v in G). Then the call $\text{DFS}(G)$ assigns $\text{post}(u) > \text{post}(v)$.

Proof. Suppose, toward a contradiction, that $\text{post}(u) < \text{post}(v)$, and split into cases:

- Either $\text{pre}(v) < \text{pre}(u)$ thus, from the parenthesis theorem, $\text{pre}(v) < \text{pre}(u) < \text{post}(u) < \text{post}(v)$, and u is a descendant of v in π . Therefore, there is a path from v to u in G . Combining this with the given that there is a path from u to v , there is a cycle in G , which contradicts G being a DAG.
- Or, $\text{pre}(v) > \text{pre}(u)$ thus, from the parenthesis theorem, $\text{pre}(u) < \text{post}(u) < \text{pre}(v) < \text{post}(v)$, and u is not an ancestor of v in π . Let w be a vertex on the path between u to v , it can be shown that if Hence, there is no path from u to v in G which contradicts the given.

□

Claim 3. Define the relation $v \sim u$ to hold if u and v are on the same strong connected components. Then:

1. $\sim$ is an equivalence relation.
2. Define the graph $G' = (V', E')$ such that V' are the equivalence class respecting to $\sim$ and $\{u, v\} \in E'$ if there is a directed path in G from a vertex in u into a vertex in v . Then G' is a DAG.

Claim 4. Let u the vertex with the minimal post value, then $\text{Explore}(G, u)$ colorize only the vertices on u 's strong connected components.